MATH 2B: SAMPLE MIDTERM #1

- This exam consists of 5 questions and 85 total points.
- Read the directions for each problem carefully and answer all parts of each problem.
- Please show all work needed to arrive at your solutions (unless instructed otherwise). Label graphs and define any notation used. Cross out incorrect scratch-work.
- No calculators or other forms of assistance are allowed. Do not check your cell phones during the exam.
- Clearly indicate your final answer to each problem.

1. (15 points)

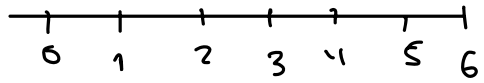
a. Estimate the area under the graph of $f(x) = x^2 + x$ from $x = 0$ to $x = 3$ using 3 approximating rectangles and left endpoints.

$\Delta x = 1, \Rightarrow 1(f(0) + f(1) + f(2))$

$= 1(0 + 2 + 6) = 8$

b. Estimate the area under the graph of $f(x) = x - 1$ from $x = 0$ to $x = 6$ using 3 rectangles and midpoints.

$\Delta x = 2$



$2(f(1) + f(3) + f(5)) = 2(0 + 2 + 4) = 12$

c. Find an expression for the area under the graph of $f(x) = x^2 + x$ from $x = 2$ to $x = 5$ as a limit of Riemann sums. (You do not need to evaluate the limit.)

$$\lim_{n \to \infty} \sum_{i=1}^{n} \frac{3}{n}\left[\left(\frac{3i}{n} + 2\right)^2 + \left(\frac{3i}{n} + 2\right)\right]$$

2. (15 points) Evaluate each of the following indefinite integrals.

a. $\int x\sqrt{3x^2-1}\,dx$

let $u = 3x^2 - 1$

$du = 6x\,dx$

$= \frac{1}{6}\int 6x\sqrt{3x^2-1}\,dx$

$= \frac{1}{6}\int \sqrt{u}\,du = \frac{1}{6}\cdot\frac{2}{3}u^{3/2} + C$

$= \frac{1}{9}\left[3x^2-1\right]^{3/2} + C$

b. $\int \frac{1-\sin^2(x)}{\cos x}\,dx$

Remember! $\sin^2(x) + \cos^2(x) = 1 \iff 1 - \sin^2 = \cos^2(x)$

$= \int \frac{\cos^2(x)}{\cos(x)}\,dx = \int \cos(x)\,dx = \sin(x) + C$

c. $\int \sin(7\theta+5)\,d\theta$

let $u = 7\theta + 5$

$du = 7\,d\theta$

$\frac{1}{7}\int 7\sin(7\theta+5)\,d\theta$

$= \frac{1}{7}\int \sin(u)\,du = -\frac{1}{7}\cos(u) + C$

$= -\frac{1}{7}\cos(7\theta+5) + C$

3. (15 points)

a. Find the average value of the function $f(x) = \tan^3(x)\sec^2(x)$ on the interval $[0, \frac{\pi}{4}]$.

(not on midterm ...) formula is $f_{avg} = \frac{1}{b-a}\int_a^b f(x)\,dx$

$$\frac{1}{\frac{\pi}{4}-0}\int_0^{\pi/4} \tan^3 \sec^2(x)\,dx$$

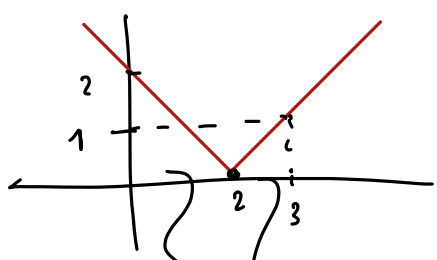
let $u = \tan(x)$ $\quad du = \sec^2 x\,dx$

x	u
0	0
$\pi/4$	1

$$= \frac{4}{\pi}\int_0^1 u^3\,du = \frac{\cancel{4}}{\pi}\left[\frac{u^4}{\cancel{4}}\right]_0^1$$

$$\boxed{\frac{1}{\pi}}$$

b. A particle moves along a line so that its velocity at time t is $v(t) = |2-t|$. Find the displacement of the particle during the time period $0 \le t \le 3$.



(Draw graph, easy to find by geometry)

$$\underbrace{\tfrac{1}{2}(2)(2)}_{2} + \underbrace{\tfrac{1}{2}(1)(1)}_{\frac{1}{2}} = 5/2$$

By integration: $|2-t| = \begin{cases} 2-t & t \le 2 \\ t-2 & t \ge 2 \end{cases}$

$$= \int_0^2 2-t\,dt + \int_2^3 t-2\,dt$$

$$\left[2t - \frac{t^2}{2}\right]_0^2 + \left[\frac{t^2}{2} - 2t\right]_2^3 = [4-2] + \left[\frac{9}{2} - 6 - \frac{4}{2} + 4\right]$$

$$= 2 + \frac{1}{2} = 5/2$$

4. (20 points)

a. Complete the blanks in the following statement of the Fundamental Theorem of Calculus.

Fundamental Theorem of Calculus:
Suppose f is continuous on $[a, b]$. If $g(x) = \int_a^x f(t)\,dt$, then $g'(x) =$ $f(x)$ and $\int_a^b f(x)\,dx =$ $F(b) - F(a)$, where F is any antiderivative of f.

b. Use the Fundamental Theorem of Calculus to evaluate the following.

i. $\dfrac{d}{dy}\displaystyle\int_2^y \frac{\sin(t)}{t^2+3}\,dt$ $= \dfrac{\sin(y)}{y^2+3}$

ii. $\dfrac{d}{dx}\displaystyle\int_x^{x^4} \sqrt{t}\,dt$

$\sqrt{x^4} \cdot 4x - \sqrt{x} \cdot 1$

$= x^2 \cdot 4x - \sqrt{x}$

$= 4x^3 - \sqrt{x}$

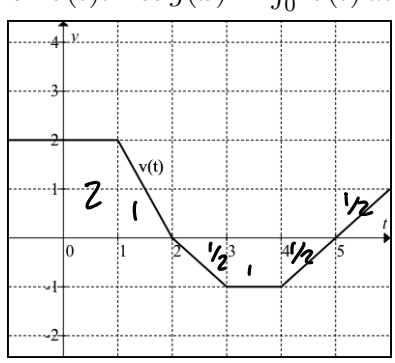
c. Answer each of the following questions. No work or explanation is needed.

i. If $f(t)$ is measured in dollars per year and t in years, what are the units of $\int_0^{10} f(t)\,dt$? dollars.

ii. True/False: All continuous functions have derivatives. F

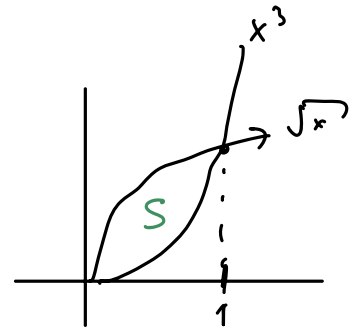
iii. True/False: All continuous functions have antiderivatives. T

iv. Below is the graph of a function $v(t)$. Let $g(x) = \int_0^x v(t)\,dt$.



Find each of the following:

$g(0) =$ 0, $g(2) =$ 3, $g'(1) =$ 2, $g'(4) =$ -1



5. (20 points) Let S be the region bounded by $y = x^3$ and $y = \sqrt{x}$.

a. Find the area of the region S.

$$\int_0^1 \sqrt{x} - x^3 \, dx = \frac{2}{3}x^{3/2} - \frac{x^4}{4}\Big|_0^1$$

$$= \frac{2}{3} - \frac{1}{4} = \frac{5}{12}$$

b. i. Find the volume of the solid obtained by revolving the region S about the x-axis.

$$\pi \int_0^1 (\sqrt{x})^2 - (x^3)^2 \, dx = \pi \int_0^1 x - x^6 \, dx$$

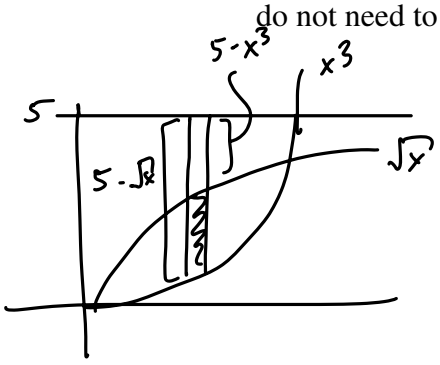
$$= \pi \left[x^2 - \frac{x^7}{7} \right]_0^1 = \pi \left(1 - \frac{1}{7} \right)$$

$$= \frac{6\pi}{7}$$

ii. Set up an integral to find the volume obtained by revolving S about the y-axis. (You do not need to evaluate the integral.) $x = \sqrt[3]{y}$ $x = y^2$

$$\pi \int_0^1 (\sqrt[3]{y})^2 - (y^2)^2 \, dy.$$

iii. Set up an integral to find the volume obtained by revolving S about the line $y = 5$. (You do not need to evaluate the integral.)



$$\pi \int_0^1 (5 - \sqrt{x})^2 - (5 - x^3)^2 \, dx$$

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